

Decimal Numbers

Look at this code:

```
distance = float(input("Enter kilometres: "))
travel_minutes = float(input("Enter travel minutes (not 0): "))

speed = distance / travel_minutes

print("Kilometres per minute:", speed)
```

Use the code above to help you. After completing each challenge below, run your program to test that it works correctly. Save each completed challenge program in a new file.

Programming Challenges

Challenge 1

Change the example so it also divides `distance` by 2 and stores the result in `halfway_distance`. Display "Halfway distance:" followed by this value before the speed. For example, 9 and 3 display "Halfway distance: 4.5".

Challenge 2

Extend the example so it asks for rest minutes, converts the answer with `float()` and stores it in `rest_minutes`. Add `travel_minutes` and `rest_minutes` and display "Total minutes:" followed by the result. For example, 10, 12.5 and 2.5 display "Total minutes: 15.0".

Challenge 3

Create a carpet calculator. Ask for a room's length, width and the carpet price per square metre, converting each answer with `float()`. Display "Floor area:" and "Carpet cost:" with the results. For example, 4.5, 2 and 10.5 display "Floor area: 9.0" and "Carpet cost: 94.5".

